Open-Source Hardware

Schematic, layout, firmware — published & freely shared

OSHWA • Licensing • Repositories • Community

Audience: Engineering Students & Makers

What • Why • Licenses • How to publish • Communities

What is Open-Source Hardware?

Definition + spirit

Source available

Schematic, PCB layout, BOM,
firmware all public.

Freedom to study

Anyone can learn how
it works.

Freedom to modify

Fork, improve, derive
new designs.

Freedom to share

Distribute changes
with the same freedoms.

Freedom to make

Anyone can manufacture
the design themselves.

OSHWA definition

Open Source Hardware Association
formal statement of OSH.

Famous Open-Source Hardware

Projects you've probably touched

Arduino

First mainstream OSH.
2005, Italy.

Raspberry Pi

Schematics public,
closed silicon.

BeagleBone

Fully open SBC.
Texas Instruments AM335x.

Olimex

Bulgarian OSH vendor.
KiCad-designed boards.

Adafruit

Hobby-focused designs.
Thousands of boards.

KiCad itself

GPL EDA tool, CERN.
Most popular OSH EDA.

Why Open-Source Your Hardware?

Reasons that go beyond ideology

- Faster ecosystem growth — third parties build accessories
- Trust — security researchers can audit the design
- Education — students learn from real designs, not toys
- Cheaper components — others volume-buy the same parts
- Longer life — community keeps designs alive after vendor exits
- Marketing — 'open source' is a recognised value signal
- Talent attraction — engineers want to work on transparent tech
- Compliance — clearer audit trail for safety / security certs

Hardware Licenses

Which one to pick

License	Type	Allows commercial use	Copyleft?
CERN-OHL-S	Hardware	Yes	Strongly (must share derivatives)
CERN-OHL-W	Hardware	Yes	Weakly (share design changes)
CERN-OHL-P	Hardware	Yes	None (permissive)
TAPR Open Hardware	Hardware	Yes	Light copyleft
MIT / BSD	Permissive	Yes	No
Apache 2.0	Permissive	Yes	No (patent grant!)
GPL / LGPL	Software (also HW)	Yes	Strong / weak copyleft
Creative Commons	Docs / images	Varies	BY-SA = share-alike

OSHWA Certification

Official mark of compliance

- OSHWA — Open Source Hardware Association
- Free certification mark for projects that meet the OSH definition
- Get a unique UID (e.g. IN000123)
- Project listed in the OSHWA registry
- Display the OSHWA logo on the PCB silkscreen
- Apply at certification.oshwa.org
- Requires: schematic, PCB, BOM, documentation, license posted publicly

Where to Host Your Hardware

Pick a home for your design files

GitHub

Most popular, free.
KiCad-friendly diff.

GitLab

Self-hostable.
Good CI for hardware.

Hackaday.io

Project-style hosting +
community engagement.

Hackster.io

Maker community,
tutorials, recipes.

CADLab

Schematic-aware
version control.

Tindie / OSHpark

Sell your design
or get it manufactured.

Documenting Open-Source Hardware

The deliverables matter

- README.md — project overview, license, build instructions
- Schematic PDF — current revision, well annotated
- Source files — .kicad_sch / .kicad_pcb (or Altium equivalents)
- Gerber + drill files — for fab houses
- Bill of Materials (CSV) — manufacturer + part number for every reference
- Pick-and-Place (.pos) file — for assembly houses
- 3D model / STEP — for enclosure designers
- Firmware repository — link to companion code
- Revision history (CHANGELOG.md) — every change documented
- Photos + videos — make it real for newcomers

Hardware-Friendly Git

KiCad + Git — best practice

- .kicad_sch / .kicad_pcb files are plain text → diff friendly
- kicad-diff plugin — visual schematic & PCB comparison
- .gitignore: backup files, cache, generated outputs
- Tag releases: v1.0, v1.1, etc. — fab fileset per tag
- GitHub Actions / GitLab CI — auto-generate Gerber on commit
- Pull requests for design review — same flow as software
- Issues for bugs, feature requests, RMA-worthy defects
- Branches for revisions: main = stable, rev-B = WIP

Sustaining an OSH Project

How to keep the project alive

- Sell assembled boards (Tindie, Amazon, your store)
- Patreon / GitHub Sponsors — direct community support
- Consulting on integration — paid hours, free design
- Donations from grateful users
- Educational programs / training courses
- Make a hardware kit — community handles soldering & support
- Plant the design in OEM products — license royalties

Hardware vs Software Open-Source

Same spirit, different problems

Software OSS

- Zero cost to copy
- Issue → patch in days
- Distribution: git push
- Licenses well-trod (GPL, MIT, Apache)
- Forking has zero cost
- Network effects (npm, PyPI) speed adoption

Hardware OSH

- Each unit costs ₹
- Issue → re-spin, re-fab → weeks
- Distribution: fab + ship physical
- Licenses less mature (CERN-OHL recent)
- Forking costs prototype $ + time
- Adoption hampered by manufacturing barriers

Common OSH Pitfalls

Mistakes new OSH projects make

- Releasing only Gerbers — not the source files (can't be modified)
- Missing license file in the repo — unclear what users can do
- No BOM with vendor part numbers — derivatives can't actually build
- Outdated documentation — schematic v1.0, repo v2.1
- Closed-source firmware — design 'open' but un-usable
- Vendor-locked components — no second source
- Ignoring issues / pull requests — community drifts away
- Trademark traps — using the name commercially anyway

Indian OSH Ecosystem

Local projects, communities, suppliers

OSHW India

Community wiki and mailing list.

FabAcademy India

Distributed makerspace network.

CDAC / IISc

OSH-adjacent research labs producing chips & PCBs.

Robu.in / Quartz

Distributors that stock breakouts + parts.

Indic chipset

Shakti, Vega — Indian RISC-V open ISA processors.

Maker Mela

Annual maker fair — Mumbai / Bengaluru.

Key Takeaways

Key takeaways from this lecture

Define your scope

What you're opening + why.

Pick a license

CERN-OHL / MIT / Apache for OSH.

Publish everything

Schematic + PCB + BOM + firmware.

Git-friendly EDA

KiCad's text files diff well.

Document obsessively

README + photos + videos.

Sustain through engagement

Replies, releases, community.

Recommended Resources

OSH learning paths

- oshwa.org — official OSH association & certification
- ohwr.org (CERN OHR) — high-quality OSH portal
- Hackaday.io — project hosting + community
- Tindie.com — sell your own hardware
- Adafruit / Sparkfun / Pimoroni — exemplary OSH practitioners
- RISC-V International — open instruction set
- TAPR Open Hardware License documentation
- Indic-specific: SHAKTI (IIT Madras), Vega (CDAC)

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- OSHWA.org — Open Source Hardware Association
- OHWR.org — CERN Open Hardware Repository
- Hackaday.io / Hackster.io — project hosting
- CERN-OHL v2 license texts (free)
- Adafruit / Sparkfun — example OSH companies
- Tindie.com — marketplace for OSH
- RISC-V — open processor ISA
- Indian: Shakti, Vega, IISc-Open Silicon

Thank you • Build something this week.